

Three recommendations to engage At-Risk Students in Critical Reflection on Intelligent Technologies through Remote Learning

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ABSTRACT

The agenda of Computational Empowerment points towards the need for inclusive approaches for supporting all students in learning about technology and the development of digital literacy. This paper aims at exploring how to develop remote learning activities in technology education during a pandemic for at-risk students. We present a case study with 23 primary students (11-12 years) who we involved in online and offline learning activities about both how intelligent technologies work but also on the implications that these technologies bring to our society. Our findings showed potential challenges to engage at-risk students in the agenda of Computational Empowerment. We propose three recommendations and future directions to scaffold at-risk students' learning about intelligent technologies in remote contexts: (1) support the development of agency through the engagement of contexts-for-action; (2) reduce the complexity of remote communication through creative and bodily engagement; (3) reflect upon forces that marginalize oneself in a digitized society.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in interaction design**.

KEYWORDS

Emerging Technology Education, Societal and ethical implications, K-12 education, Computational Empowerment, At-Risk Students, Remote Learning, Design Fiction, Critical reflection

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1 INTRODUCTION

The agenda of Computational Empowerment in K-12 education foresees the development of students' digital design literacy to promote active citizenship of all children, including marginalised youth and

minority groups [23]. This aim focuses on engaging students constructively, analytically and critically in broader questions about the impacts of technologies on their lives and society at large. The increased availability of intelligent technologies has expanded to both students' everyday lives and moves gradually into the classroom. Hence, there is an increasing trend to engage students in ethical considerations and societal impact of intelligent technologies as part of the agenda in K-12 education [15]. For instance, several scholars have highlighted the need for novel educational strategies to teach K-12 students about intelligent technologies by reflecting on aspects that go beyond the technical features [10, 16, 31, 38]. This demand calls also for teaching practices and digital tools that support marginalized students in understanding and reacting to the implications of intelligent technologies that impact them as users [30, 39, 45].

In this regard, previous work has stressed the importance of engaging a broader range of students and the need for progressive learning strategies for underrepresented target groups such as students of diverse socioeconomic and cultural backgrounds [43]. Other scholars pointed specifically to the role of the teacher as facilitator of students' learning processes about intelligent technologies. For instance, Anwar and Bascou [6] discussed the role of the educator to design socially and culturally relevant learning activities which help students understand the technological groundings behind Artificial Intelligence (AI) research, its limits and its potentials. Malinverni et al. [28] stressed the potential of educational robotics as a space to open the debate around gender, intersectionality and its impact on society. Long and Magerko [27] presented design considerations based on key concepts in learner-centered AI for developers and educators. Finally, Touretzky et al. [40] proposed guidelines that also include critical reflections on the opportunities and risks that AI applications entail.

In alignment with these academic discussions, policymakers are requesting novel educational strategies to integrate technology education in the K-12 school curriculum. For instance, the Informatics for All coalition has suggested to integrate the subject informatics into other disciplines with the aim of teaching students about significant applications of this domain so as to include their social impact and relevance to the future of work and life in general [13]. At the same time, other political organizations [2, 42] and new research agendas in general [4, 5] have addressed intelligent technologies at large, focusing on the need for explainable, transparent, ethical aspects of technological innovations. Such characteristics of intelligent technologies are difficult to teach to children because they are abstract concepts. Learning strategies in technology education should support students in critically reflecting on these characteristics and their ethical and societal implications, i.e. their

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development of Computational Empowerment in emerging technology education [31].

Previous research has highlighted the opportunities for design-based and hands-on activities to engage children in technology education [35]. Other approaches have focused specifically on the background characteristics of the students, demanding methods that more broadly consider their cultural and social context, as well as their personal previous dispositions with respect to them [6]. However, during the COVID-19 pandemic, most educational activities moved online and suffered from several challenges such as work in isolation, lack of access to school labs and equipment, aspects related to digital gaps at home [19], lack of general pedagogical resources and confidence by teachers to shift to online work [7] and lack of telework habits between students and teachers. Scholars reported also relevant challenges when engaging students in physical computing activities remotely [14]. The already present widening general educational gap between students with and without access to technology identified in the Programme for International Student Assessment [1] increased for students from low socio-economical families [20] due to the limited access to internet, lack of availability of technological devices (computers, tablets, etc.), or the capacity by parents to support online learning activities [2, 3], as well as a higher impact in economic (e.g. loss of wages or basic insecurities such as food provision), and personal (e.g. anxiety) situations.

This case study aims at exploring how to develop remote learning activities in technology education during a pandemic for at-risk students. We define at-risk students as children and teenagers who live in situations where poverty and socio-cultural norms within their families interfere with their opportunities to access educational resources and learning opportunities. In this context, we consider remote learning as educational activities where the learner and teacher communicate with a physical distance and learning contents are taught through other means than in traditional classroom settings. Our goal in this case study is to answer the following questions:

- How can we scaffold students' remote learning about intelligent technologies during a pandemic?
- Which specific educational strategies and digital tools can support at-risk students in their learning processes in remote contexts?
- How can we engage students in remote learning about the technical and ethical implications that these technologies bring to our society?

The contributions of our work are three recommendations to engage at-risk students' in remote learning about intelligent technologies: (1) support the development of agency through the engagement of contexts-for-action; (2) reduce the complexity of remote communication through creative and bodily engagement; (3) reflect upon forces that marginalize oneself in a digitized society. Our findings are based on a study with 23 primary students of a public school in Barcelona, a semi-structured interview with the class teacher, her evaluation of the students' contributions and researchers' observation notes. Our results using thematic analysis showed how the students gradually disengaged in the learning activities during the study and highlighted issues about teaching methods, the used physical materials and digital tools. To conclude,

we discuss potential challenges and possible future directions to include at-risk students through remote learning in the agenda of Computational Empowerment.

2 EDUCATIONAL STRATEGIES FOR TECHNOLOGY EDUCATION WITH STUDENTS FROM AT-RISK COMMUNITIES

Recent reports demonstrate an educational gap between students' from low socio-economical families and the general population regarding their access to technology and involvement in technology education [2]. To address these challenges, there has been an increasing demand for efforts involving marginalized students and minorities in computing activities. Several scholars have focused on understanding social, cultural, individual and contextual aspects that support students' participation in diversity contexts. For instance, Schepers et al. [32] highlighted the importance of developing self-esteem, learning-by-doing, and broadening horizons when involving children at risk of marginalization in digital fabrication workshops. Kafai et al., [24] stressed the need for allowing students personalized pathways into computing in order to engage individuals who might otherwise be excluded from the field. Similar, in earlier work, Kafai et al. [25] argued that open-ended design experiences can cultivate learner agency while also developing learners' identity within their cultural context and place. In this regard, Roque et al. [29] particularly pointed out how embedded biases in creative technologies can impact "what" and "how" the students can express their ideas. Specifically, their findings highlighted that families from latino-american backgrounds could not represent themselves and their cultural context using the physical computing tool Scratch Junior. Iivari et al. [21] pointed towards how teaching methods can exclude students from engaging in the learning activities in technology education. Braun et al. [11] argued for the importance of contextual dimensions that can influence teaching practices such as situated, professional, external and material contexts.

2.1 Learning about societal and ethical implications of intelligent technologies

Several researchers have stressed the importance of teaching practices that support marginalized students in understanding and reacting to forces that marginalize them. For instance, Yu et al. [45] stressed the importance to train participants in the language skills and vocabulary necessary to communicate social issues in narratives during storytelling activities. The aim of their study was to enable the students to tell their stories and issues they care about, e.g. sociopolitical problems within their own community. Tissenbaum et al. [39] formulated the need for "means of personal and social creative expression", e.g. by engaging students in critical reflection on inequitable societies, unsustainable energy practices and systems of oppression. Schaper et al. [30] showed that students in high socio-cultural diverse contexts can draw from their own experiences and connect them to multiple perspectives when discussions on ethical and societal issues around intelligent technologies are situated within their cultural practices.

One promising approach to teach students in diversity context about technologies has been design fiction and speculative design.

In general, this approach has been suggested as a way to speculate about new ideas through prototyping and storytelling [37]. Sharma et al. [34] highlighted how this approach enables to teach children about diversity and inclusion and helps "them to advocate such important values in their future life". This method has also been used in K-12 education in regular classroom settings and online contexts. Lane & Solis [26] used science fiction prototyping to support the development of digital literacy with elementary students. Soto et al. [36] used a science fiction narrative to engage 8th grade students in discussing environmental and sustainability topics. Wargo & Alvarado [44] proposed speculative worldbuilding to support early literacy and opportunities for young children to reflect on present realities. Tamashiro et al. [38] used a fictitious storyline and role-play activities in an online workshop to support students in the reflection upon the different interests and stakeholders in relation to facial recognition technologies. This approach gave the students an opportunity to adopt different perspectives and gain an understanding of the social, economic, and ethical values and dilemmas of these technologies. In this regard, Iivari et al. [22] warned about the ethical aspects that need to be contemplated when engaging children in speculative design activities and to provide "cultural critique, provoke, and shock, and question cultural norms and rules". The consideration of ethical aspects can be particularly relevant in context with a high diversity of cultural backgrounds and requires a sensitive facilitation approach by involved educators.

Building on previous work, we developed together with a primary teacher learning activities focusing on hands-on activities, design-oriented learning and speculative design for primary students from at-risk-communities. In the following section, we present a detailed procedure of the five weeks intervention, data collection, analysis and findings. Finally, we discuss potential recommendations and future directions for remote learning activities in these contexts.

3 CASE STUDY

This case study is part of an interdisciplinary research project that involves developing new educational practices aimed at balancing technological and societal aspects of intelligent technologies, and engaging students in abstract thinking, hands-on construction and redesign activities. Specifically, in this case study, we aim at addressing the aspect of diversity in technology education, i.e. to engage students from at-risk communities in learning about intelligent technologies. These students are often affected by the "digital divide". In other words, these students live in low socio-economic contexts where they have less access to new technologies than the rest of the population. To compensate for this gap during the COVID-19 pandemic, the schools in Barcelona offered families without computers at home to borrow the equipment from the school. However, this did not include the setup and maintenance of the equipment which could be an issue for parents with less technical knowledge and experience with technology.

4 PROCEDURE

The case study was carried out on five consecutive weeks during the lockdown in Spain due to the COVID-19 pandemic in the year 2020. 23 primary school students (girls = 11; boys = 12) between

11 and 12 years participated in the activities. The families of the participating children came from eleven different countries: Spain (1 student), Philippines (5 students), Peru (1 student), Brazil (1 student), Bolivia (1 student), Dominican Republic (2 students), India (2 students), Morocco (4 students), Pakistan (2 students), Bangladesh (2 students) and Algeria (2 students). Before the study, the responsible teacher sent the project information and the informed consent to legal guardians. In the consent forms we asked them and the students for their agreement on data collection and dissemination. The parents were also invited to get in contact with our research team for further information. The students lived with their families in El Raval, a district in the heart of Barcelona known for its high level of immigration and cultural diversity. The same students had participated a few months before the start of the global pandemic COVID-19, in an educational activity with the aim to support their reflections upon ethical issues in robotics [30]. The goal of the present case study was to build upon their previous learning experiences in robotics and extend their reflections on the implications of intelligent technologies in contexts that are part of students' everyday life. The overall theme of the workshop series was to design objects and intelligent agents that could be used at home, at work or in public spaces in a near future, i.e. after the end of the lockdown in Spain. At that time, politicians and scientists announced that we would coexist with the restrictions of the coronavirus and new life-routines during an indefinite period of time. This new situation posed two main speculative questions: *What does this new reality mean for our personal life and coexistence with our family and other people in society? How can technology help us to manage these changes?*

To explore these questions with the students, we focused on design-oriented learning activities such as speculative prototyping and computational computing activities. The activities were facilitated online and offline using virtual teaching tools. The overall process of the workshop followed the five steps of the D.School design thinking method [41]: (1) empathize, (2) define, (3) ideate, (4) prototype and (5) test. We co-developed the learning activities together with the responsible class teacher. Furthermore, the didactics were specifically adapted to remote learning methods and the fact that the children were confined at home with their families. At the beginning of each of the five weeks, the teacher published in the school blog webpage one activity that included the main objectives, a step-wise procedure, a list of deliverables and inspirational examples. Some of the activities involved the participation of other family members as a way to broaden children's perspective and speculative imaginaries on the theme. The students were invited to attend two online classes per week of a duration of 60 min each. In the first weekly class (Monday), we presented the activities for the week and the students had the opportunity to ask clarifying questions. In the second weekly class (Wednesday), the students gave an update of their process and they could reach out for help if they had difficulties to continue with the activity. During the online classes two teachers and at least one researcher were present. In the following sections, we will describe each weekly activity in detail.

Week 1

The goal of the first week was threefold: (1) to empathize with the design context which was the time after the first lock-down of the COVID-19 in Spain; (2) to define one relevant personal or

societal issue that could affect people during a future scenario of the COVID-19 pandemic; (3) to define a design idea of an intelligent technology that would address this issue. To this end, the students were first instructed to list all COVID-19 related restrictions that were affecting their daily lives, their family or other people at that time. In a next step, they were encouraged to reflect upon and note down which of these restrictions would continue in the coming months, if they had heard of any new restrictions coming up in the future and how these could affect people. To further investigate this part, they interviewed at least one family member and complemented their list with additional information. Subsequently, they chose one of the most relevant issues and brainstormed ideas for a robot or intelligent object that could help them in one of these situations. The final idea was documented as a brief explanation and as a drawing that illustrated its functioning. Finally, the students took a photo of the list, the description and the drawing and uploaded the documents to the school blog. The work-in-progress ideas were discussed during the following online class.

Week 2

The goal of the second week was to iteratively ideate and prototype design ideas of intelligent technologies. First, the students were encouraged to remember the feedback that they had received during the online class when they presented their work-in-progress ideas to the teachers and their peers. They documented this feedback on a paper. Having this feedback in mind, they transformed then their drawing into a 3D-prototype using e.g. recycling materials, cardboard, play doh, lego bricks, bottles, fabrics or any other material available in their homes. They documented the first prototype by taking a photo. Subsequently, they presented their idea and prototype to at least one family member and asked for their feedback, which they documented again in a written format. Based on these comments, the students made improvements of the prototype and took a second photo. They wrote a brief explanation of how the intelligent technology worked. Finally, the students uploaded the feedback notes, the two photos of the prototype and the brief explanation to the school blog and presented the outcomes in the next online class.

Week 3

The goal of the third week was to ideate the functioning of the intelligent technologies with the physical computing tool micro:bit. Since the students did not have the physical computing boards available at home, we used the micro:bit online platform (<https://makecode.microbit.org>) which allowed to program and simulate the behavior of LEDs, buttons and sensors. Before the lockdown the students had participated in basic hands-on activities with the physical tool and we assumed that they were familiar with its use. For this activity, the students were first instructed to revise all notes and ideas that they had documented during the first two weeks. In a next step, they were asked to imagine and document: (1) three aspects that they wanted to teach their intelligent technology; (2) three aspects that they wanted it to do with them [16]. Furthermore, we provided them with video tutorials that introduced them with the basic functionalities of the tool micro:bit and asked them to write down ideas about how they could use sensors (e.g. gyroscope, proximity sensors, etc.) and/or LEDs to simulate the behaviours of their digital prototypes. Subsequently, the students were encouraged to explain to one family member what they had learnt about

the tool micro:bit, how it could be used to program the functioning of their intelligent technology and if they had other ideas how to use the features of the micro:bit. Finally, the students uploaded the description of their inventions to the school blog. During the next online class, we revised their ideas and did some practical hands-on activities to familiarize them more with the online platform.

Week 4

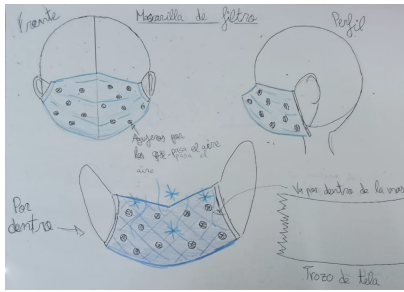
The goal of the fourth week was twofold: (1) to prototype the functioning of the intelligent technologies with the physical computing tool micro:bit; (2) to reflect upon the societal implications (accountability, biases and privacy) of their inventions. Therefore, they first implemented one of their ideas for the technological aspects of the intelligent technology using the micro:bit online platform. For this activity, we provided them with additional video tutorials personalized to their needs. In addition, the online classes were used to support the students during the programming process and to solve technical issues. Second, we asked the students to step back from the design process and think about the societal implications of their invention. These reflections focused specifically on questions around the accountability of the designer for the invention in relation to their cultural values. For instance, which beliefs and values based on their family culture and social environment (e.g. habits, traditions, religion, gender role, etc.) had influenced the behavior of their intelligent technology. We also asked them to pay attention to the implications of privacy, i.e. what types of data did their intelligent technologies collect and what would happen if someone else would get hold of this private data. The students presented their digital prototype and asked the same questions related to the societal impacts of their invention to one family member. Based on their reflections and the family members' answers, they made a list of positive and negative effects that their invention may have for society. Finally, the students were encouraged to upload the link of their micro:bit project and their list to the school blog.

Week 5

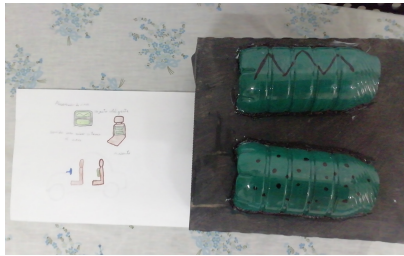
The goal of the fifth week was to document the design process of their invention. We provided them with guidelines on how to prepare a powerpoint presentation and a 3-min video describing their experience during the project. Finally, the students were asked to upload the presentation and the video to the school blog. During the online classes, the students could show their presentation in process and we gave feedback for improvements.

5 DATA COLLECTION AND ANALYSIS

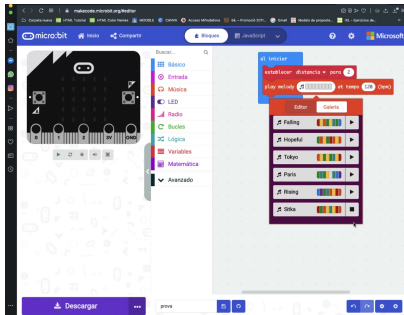
We collected qualitative data based on students' contributions that were uploaded to the school blog and researchers observation notes about each of the five sessions. Furthermore, we conducted a semi-structured interview with the responsible class teacher after the last session. The interview was video recorded and transcribed. In addition, we asked the teacher to provide us with a qualitative evaluation about the participation level and motivation of each individual student during the different activities. The first author used thematic analysis as described by Braun and Clarke [12] to interpret the qualitative data. First, she transcribed the video recordings, read through the transcripts and noted down initial ideas. Second, she discussed the initial ideas with the second author, and then coded all transcripts by applying one or more inductive conceptual



(a) Sketch of an intelligent face mask that regulates its inside-temperature



(b) Prototype of an intelligent COVID-19 detector through temperature sensors



(c) Micro:Bit online platform

Figure 1: Examples of students' design process

categories (e.g. issues that prevented the students from taking fully part in the activities) to discrete parts of the data, usually one or a few sentences. This was an iterative process and in an ongoing dialogue with the second author. Thirdly, the first and second author collated categories into potential themes. Lastly, she produced a report of the analysis illustrated with compelling quotes from the participants, and related back the analysis to the research questions.

6 RESULTS

In this section, we report on the results based on students' contributions during the five weeks of the case study, a semi-structured interview with the class teacher, students' evaluations and researchers observation notes. We focused on three core aspects: (1) students' engagement and participation during the different activities; (2) interactions with family members and peers during the design process; (3) the teachers self-perception on possible issues related to the learning activities.

Students' engagement and participation during the different activities

13 of 23 students ($f=7;m=6$) proposed ideas for robots and intelligent objects. During the ideation phase, the students looked for similarities of real objects in their closer surroundings. In other words, students' ideas were focused on robotizing existing objects or actions. Their ideas were related to (1) services in public spaces, e.g. an intelligent vacuum cleaner to collect used face-masks in the park; (2) medical services, e.g. an intelligent stretcher that cares for and operates sick people; an intelligent COVID-19 detector (see Fig. 1b) and (3) personal needs, e.g. a robot clown that cheers up people who are sick or an intelligent face mask that regulates its inside-temperature (see Fig. 1a). 11 of these students produced prototypes of their idea with low-tech materials and investigated possible implementation strategies. For instance, for the intelligent face mask idea, the girl proposed to use the temperature sensor. Another child wanted to use the icons on the micro:bit display to express emotions with the clown robot. To simulate the functioning of an intelligent vacuum cleaner, a third child would use the distant sensor to detect the waste, the display to indicate when the waste storage was full and the function to count steps for the navigation of the robot (see Fig. 1c).

However, only 6 students managed to simulate their ideas on the virtual micro:bit platform. All of these students participated in the weekly online tutorial sessions. Since they were divided into smaller groups, the students were able to experiment with micro:bit under the online supervision of the teachers and the researchers. One student was able to complete the entire programming process on his own. He designed an intelligent stretcher and had prior knowledge of the programming language python. The other students expressed, despite having video tutorials and individual support available, that they felt unsure about how to use the micro:bit online platform.

Only one student finished the entire project with a reflection upon the ethical and societal issues on her invention and documenting the design process in a powerpoint presentation. This girl designed an intelligent robot that collects and refills empty gel dispensers. In the deliverable of the fourth week, she described that she had chosen this robot because they had an ordinary gel dispenser at home and it inspired her for the invention. In relation to her reflection upon ethical and societal she described: *"It collects different types of disinfection gel dispensers. It would not matter if someone has access to this data because there isn't anything to hide."* Since the other students did not upload this deliverable, we specifically focused in our weekly online tutorial session on discussions about the ethical and societal issues of students' inventions. Overall, we could observe the same insights, i.e. the students did not have sufficient knowledge about possible potentials and risks of intelligent technologies in order to create a deeper debate about these topics. Instead, we explained the students which ethical considerations, for instance, face-recognition technologies or robots for remote surgeries, may entail. The students reacted with surprise but still had difficulties to understand how these implications may be relevant for their own everyday lives.

Interactions with family members and peers during the design process

The involvement of family members in the activities was successful only to some extent, i.e. they helped their children with manual

tasks to create the prototype or with the ideation of the ideas. In contrast, they were not able to support their children in the more technical tasks of the activities. In addition, the teacher explained in the interview that several parents had difficulties with the language, i.e. did not speak or understand Catalan nor Spanish. Also, it seemed that they could not stimulate students' reflections on ethical and societal implications on their inventions. For instance, the same girl described above who completed the entire design process reported about the feedback of their family: *"They told me that it would not matter if someone would access the data of the robot because if its task is related to hygiene, it would only collect things related to this topic."* In other words, also the family members were not aware of possible implications of intelligent technologies, i.e. that the robot may collect data about the frequency or amount of gel that people may use from the dispenser and how this could inform predictions about behavioural patterns of collectives and individuals.

Teachers self-perception on possible issues related to the learning activities

The class teachers confirmed many of these findings presented above. During the evaluation of the students, she explained about one of her strong students: *"From the beginning, she thought it would be a good idea to do the robot project. She immediately started thinking about her project, but once she got to the part of creating the prototype she saw that it was more complicated than she had thought and she didn't finish it. She told us that she was trying to do the project with the micro:bit but she didn't send it to us. Although she liked the project she did not put 100 percent of her abilities."* Most of the students faced the same problems. They were attentive to the tasks that were published each week on the school blog and were very engaged in the beginning of the project. According to the teacher, one major obstacle for the project was that the students were not used to working on project-based tasks autonomously. Usually, in a normal class situation before the pandemic, they were guided by the teacher or asked their peers for help. The few students who finished their project on the online micro:bit platform participated in the online tutorial classes in which two teachers and one researcher were present. In the tutorial classes, the students received individual support for their projects. Interestingly, one student only focused on the technical part of the project. The teacher explained: *"He showed great interest in being able to do his project with the micro:bit and, until he achieved what he had imagined, he did not stop. He had the excuse to be able to also learn to program with JAVA and after finishing the project he still told us what he was practicing. He just sent us the micro:bit project, he didn't even make a design or a model."* In this regard, the teacher highlighted that the student had always been very interested in programming and technical school activities. In the interview, the teacher suggested that future iterations of the prototype may be improved by creating a "hyper-controlled" context of a compulsory course and guide the students through every small detail of the tasks.

Overall, our findings point towards several challenges that the students faced related to the digital tools used during the design process, the overall teaching method, collaboration with peers and active involvement of family members in technical and reflection

tasks. In the next section, we will discuss how to tackle these challenges and provide possible entry points for supporting at-risk students in remote learning.

6.1 Discussion

In this study, we explored how to engage a primary school class of 23 at-risk students remotely in learning about intelligent technologies during the strict COVID-19 lock-down in Spain. We focused on the concept of Computational Empowerment [23] that points towards the need for inclusive approaches for supporting all students in learning about the technical aspects but also about the impact and implications of intelligent technologies for society at large. Our findings showed that (1) only half of the students engaged with the proposed activities and that their participation gradually decreased with the complexity level of the tasks; (2) activities that required the involvement of family members did not lead to insightful discussions nor informed children's design process of an intelligent technology; (3) the techniques that we used hardly triggered discussions about ethical and societal implications of intelligent technologies. Based on the identified challenges, we developed three design recommendations for learning activities and tools in the context of remote learning about intelligent technologies. We will now outline these three recommendations and point towards future directions for exploring novel teaching strategies for at-risk student in these contexts.

Design Recommendation 1: Support the development of agency through the engagement of contexts-for-action

Our outcomes showed that half of the class produced prototypes of their ideas with low-tech materials and investigated possible implementation strategies for the technological aspects. Yet, only one third of the class actually managed to implement at least one of their ideas with the online version of micro:bit. As a possible explanation, the teachers pointed out that the students had less individual support during the activities than in the physical classroom setting. Further, the teacher suggested that students' participation may be increased if students were required to complete the tasks in the context of a compulsory course. In this regard, Iivari et al. [21] highlighted that the exclusion of students in technology education can be created by the teaching method choices. In the context of our study, we believe that the lack of students' participation may not be solved by providing a compulsory activity and more "controlled" environment, instead students should be enabled to take agency through the active engagement of individuals with aspects of their contexts-for-action [9]. In other words, the transformation of teaching practices can not only be reduced to the introduction of new digital tools [17] but requires creative agentive actions and collaborative co-configuration processes [18]. Possible future directions for supporting the development of students' agency may be providing digital tools in combination with physical materials that allow personal pathways [24] and open-ended design experiences [25]. These strategies have been suggested as a means to cultivate learner agency while also developing learners' identity within their cultural context and place.

Design Recommendations 2: Reduce the complexity of remote communication through creative and bodily engagement

Despite ongoing necessary discussions and initiatives to reduce the inequality of technology access of at-risk communities [2], it is important to consider the situated realities that these students are facing [11]. In our study, many families did not have internet access nor computers at home. Instead, they connected to the internet with mobile hotspots and ipads provided by the school. Furthermore, the students who participated in our study mainly finalized the hands-on activities but less those tasks that required the use of digital tools (e.g. micro:bit, recording a video or preparing a power point presentation). At the same time, the students only received help from their families during manual tasks (e.g. cutting materials, gluing the physical prototype together, etc.). To tackle challenges of remote communication, Beuthel et al. [8] proposed a series of activities to support playful, material-driven, bodily ways of remote connectedness such as making an artifact for each other, physical engagement and sharing one's experience (e.g. a bracelet or letter that is physically sent to the other person). Similar strategies could be integrated in future studies with the aim to support the interaction between family members or classmates that live at a distance. Beuthel et al. [8] also highlighted how their strategies can reduce the complexity in communication which do not "necessarily require the design of interactive, digital or hybrid means". The latter aspect may be particularly relevant for students from at-risk communities which often suffer from poverty and live in low-income families, i.e. do not have access to and knowledge about certain technologies. In other words, using other means than digital tools may allow students to creatively use materials at-hand, bodily engage with one's another and hence reduce the complexity of communication.

Design Recommendation 3: Reflect upon forces that marginalize oneself in a digitized society

An essential aspect of the development of Computational Empowerment is to support students' understanding about ethical and societal implications of intelligent technologies. In our study, we asked the students to reflect up on ethical aspects, i.e. the accountability of the designer for their invention and privacy issues in relation to their socio-cultural values. In contrast to previous studies with the same students [30], this time we did not achieve to create a deep debate on ethical issues related to intelligent technologies and specifically about students' inventions. We believe that this issue may be caused because many of students' inventions were not "close" enough to their everyday experience with intelligent technologies [31]. In other words, the learning activities did not provide them with "thick authenticity", i.e. activities that are personally meaningful to the students and link to important and interesting aspects of the world beyond the classroom [33]. In this regard, further investigation is needed to understand families' role in the ideation and formulation of students' inventions. Specifically, family members may be more actively involved in the exploration and reflection upon the use of intelligent technologies in their household (e.g. google, app with face-recognition systems, chatbots, etc.). At the same time, the framing of our speculative design scenario may have not been sufficiently supportive to stimulating students' imagination about alternative ideas of intelligent technologies. In future work, we will explore strategies to engage

students' in redesign activities for imagining alternative futures [38] and design critique of the "status quo" [22]. Furthermore, other researchers have stressed the importance of teaching practices that support marginalized students in understanding and reacting to forces that marginalize them. For instance, Yu et al. [45] stressed the importance to train participants in the language skills and vocabulary necessary to communicate social issues. Tissenbaum et al. [39] formulated the need for "means of personal and social creative expression", e.g. by engaging students in critical reflection on inequitable societies, unsustainable energy practices and systems of oppression. In other words, one important requirement for at-risk students to understand the implications of intelligent technologies may be to engage them in themes that support their reflections on the forces that marginalize themselves as an individual and consequently as an at-risk community in a digitized society.

6.2 Limitations

One main limitation of our study was that we were not able to collect data on students' self-perception about the learning activities. In the last week of the project, we had asked the children to record a short video describing their experience during the project. However, none of the students completed this task. This shortcoming suggests the exploration of other formats for online contexts and with this target group, e.g. semi-structured interviews in smaller break-out rooms. Another limitation is that we did not involve the families of the students before the study. A virtual meeting explaining our objectives, introducing them to the digital tools that we would use and an introduction of potentials and risks of intelligent technologies may have allowed them to engage more in the shared learning activities and understand the relevance of the topic for their children's education. We invite scholars in the CHI community to consider the challenges and possible future directions pointed out in this case study with the aim to engage students' from at risk-communities in the agenda of Computational Empowerment.

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